

การควบคุมระยะไกลในอุตสาหกรรมและการประมวลผลภาพ

Industrial remote control and image processing

Experiment 6/9: IRIVE Pi Control (3/3) -- Machine Vision

6.1 Read More

6.2 เนื้อหาที่เกี่ยวข้อง

6.2.1 OpenCV

6.2.2 YOLO

6.3 การทดลอง -- Machine Vision

6.3.0 -- Install OpenCV

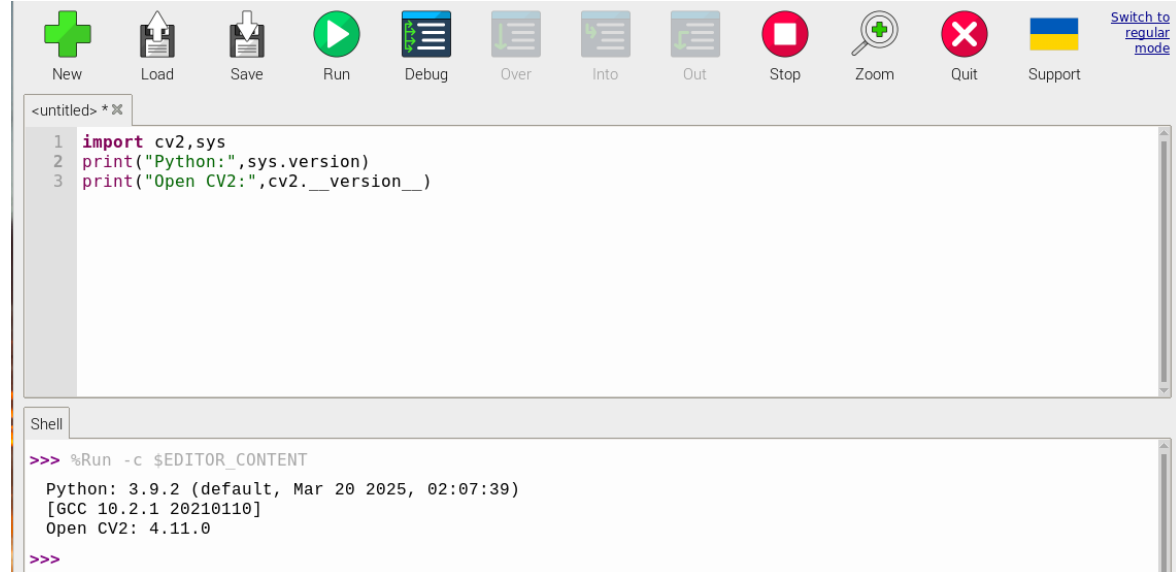
sudo apt update && sudo apt upgrade -y

```
pi@iriv:~ $ sudo apt update && sudo apt upgrade -y
Hit:1 http://deb.debian.org/debian bullseye InRelease
Hit:2 http://deb.debian.org/debian bullseye-updates InRelease
Get:3 http://security.debian.org/debian-security bullseye-security InRelease [27.2 kB]
Hit:4 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:5 http://archive.raspberrypi.org/debian bullseye InRelease
Fetched 27.2 kB in 2s (16.4 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

pip3 install opencv-contrib-python

```
pi@iriv:~ $ pip3 install opencv-contrib-python
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting opencv-contrib-python
  Downloading opencv_contrib_python-4.11.0.86-cp37-abi3-manylinux_2_17_aarch64.manylinux2013_aarch64.whl (47.9 MB)
    | 47.9 MB 86 kB/s
Requirement already satisfied: numpy>=1.17.3 in /usr/lib/python3/dist-packages (from opencv-contrib-python)
Installing collected packages: opencv-contrib-python
Successfully installed opencv-contrib-python-4.11.0.86
```

Check → Python, OpenCV2



The screenshot shows a code editor window titled "<untitled> * %". The editor contains a Python script with three lines of code:

```
1 import cv2,sys
2 print("Python:",sys.version)
3 print("Open CV2:",cv2.__version__)
```

Below the code editor is a shell window labeled "Shell". It displays the output of running the script:

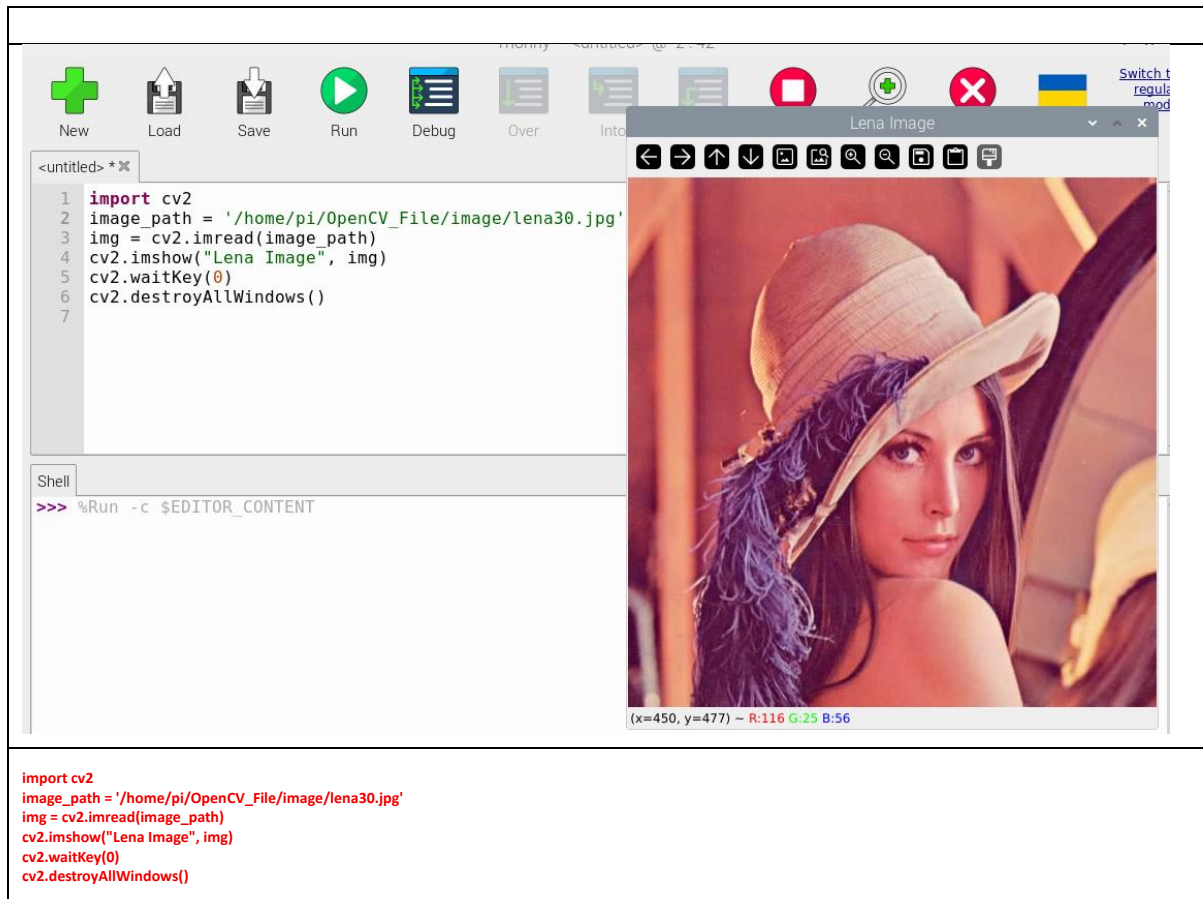
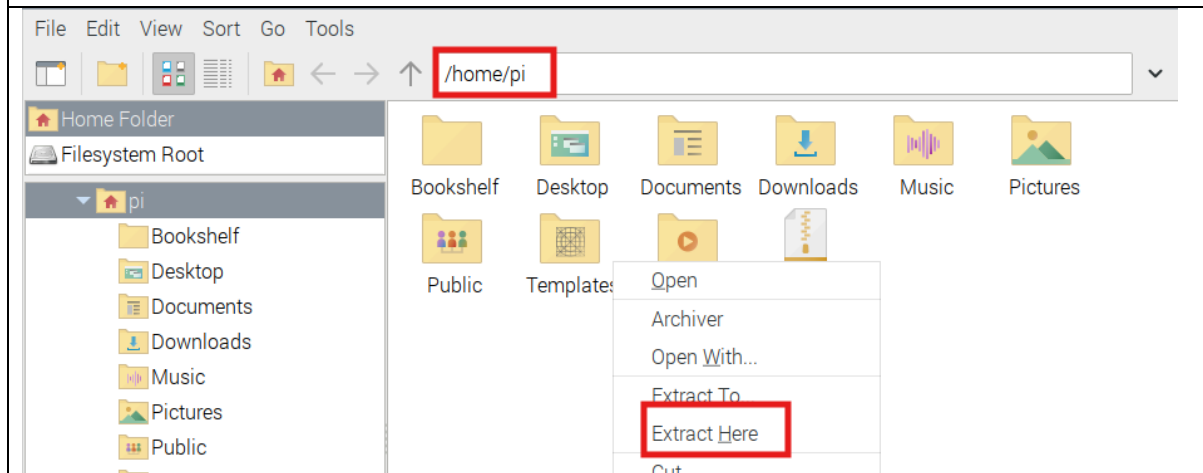
```
>>> %Run -c $EDITOR_CONTENT
Python: 3.9.2 (default, Mar 20 2025, 02:07:39)
[GCC 10.2.1 20210110]
Open CV2: 4.11.0
>>>
```

```
import cv2,sys
print("Python:",sys.version)
print("Open CV2:",cv2.__version__)
```

6.3.1 -- Copy Data File and Open Image

Copy OpenCV_File.zip to /home/pi

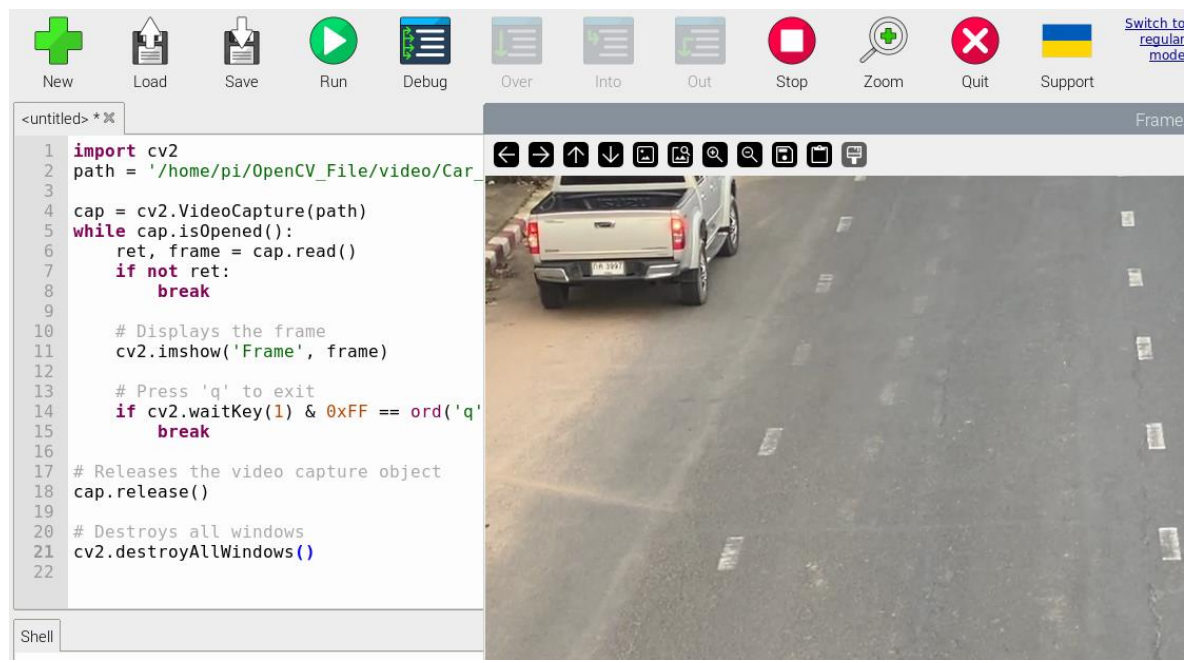
>> Extrack to Here



```
import cv2
image_path = '/home/pi/OpenCV_File/image/lena30.jpg'
img = cv2.imread(image_path)
cv2.imshow("Lena Image", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

6.3.2 -- Open Video

6.3.2.1 Open Video File



The screenshot shows the OpenCV Python IDE interface. At the top is a toolbar with icons for New, Load, Save, Run, Debug, Over, Into, Out, Stop, Zoom, Quit, and Support. Below the toolbar is a code editor on the left and a video player on the right. The code editor contains the following Python code:

```

1 import cv2
2 path = '/home/pi/OpenCV_File/video/Car_FangMaster.mp4'
3
4 cap = cv2.VideoCapture(path)
5 while cap.isOpened():
6     ret, frame = cap.read()
7     if not ret:
8         break
9
10    # Displays the frame
11    cv2.imshow('Frame', frame)
12
13    # Press 'q' to exit
14    if cv2.waitKey(1) & 0xFF == ord('q'):
15        break
16
17    # Releases the video capture object
18    cap.release()
19
20    # Destroys all windows
21    cv2.destroyAllWindows()
22

```

The video player on the right shows a frame of a silver pickup truck driving on a road. The video player has a toolbar with navigation and zoom controls. The title bar of the video player says 'Frame'.

Below the code editor is a shell window with the following code:

```

import cv2
path = '/home/pi/OpenCV_File/video/Car_FangMaster.mp4'

cap = cv2.VideoCapture(path)
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    # Displays the frame
    cv2.imshow('Frame', frame)

    # Press 'q' to exit
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

    # Releases the video capture object
    cap.release()

    # Destroys all windows
    cv2.destroyAllWindows()

```

6.3.2.2a Open USB Camera – with ffplay

**sudo apt-get update && sudo apt-get upgrade -y
v4l2-ctl --list-devices**

```
File Edit Tabs Help
pi@iriv:~$ v4l2-ctl --list-devices

bcm2835-codec-decode (platform:bcm2835-codec):
/dev/video10
/dev/video11
/dev/video12
/dev/video18
/dev/video31
/dev/media1

bcm2835-isp (platform:bcm2835-isp):
/dev/video13
/dev/video14
/dev/video15
/dev/video16
/dev/video20
/dev/video21
/dev/video22
/dev/video23
/dev/media2
/dev/media3

rpiivid (platform:rpiivid):
/dev/video19
/dev/media0

UC70: UC70 (usb-fe980000.usb-1.3):
/dev/video0
/dev/video1
/dev/media4

pi@iriv:~$
```

**sudo apt install ffmpeg
ffplay /dev/video0**

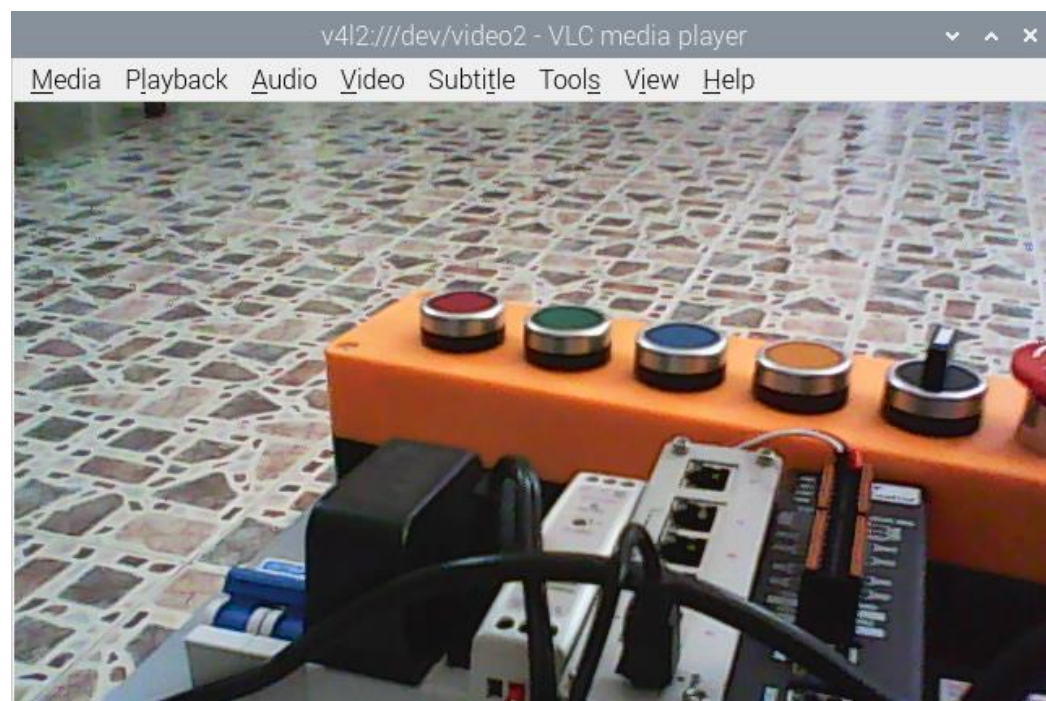
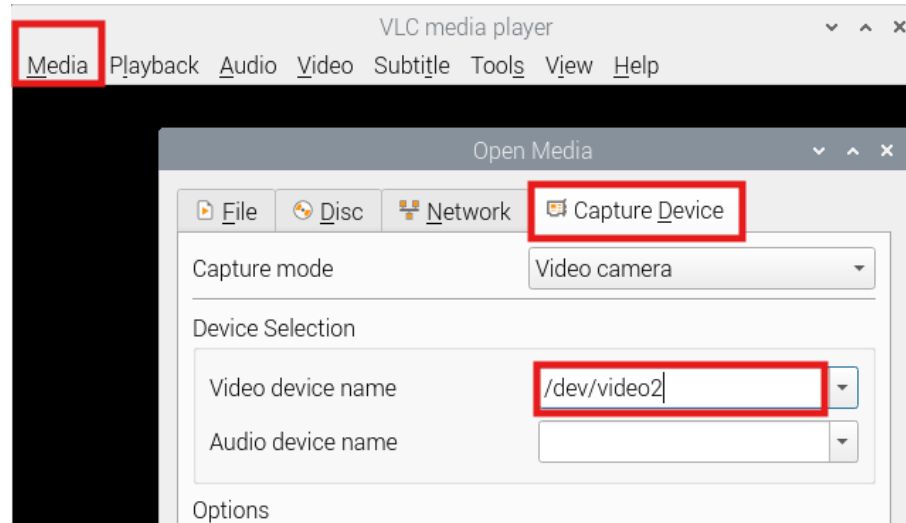
```
File Edit Tabs Help
/dev/video3
/dev/media5

UC70: UC70 (usb-fe980000.usb-1.3):
/dev/video0
/dev/video1
/dev/media4

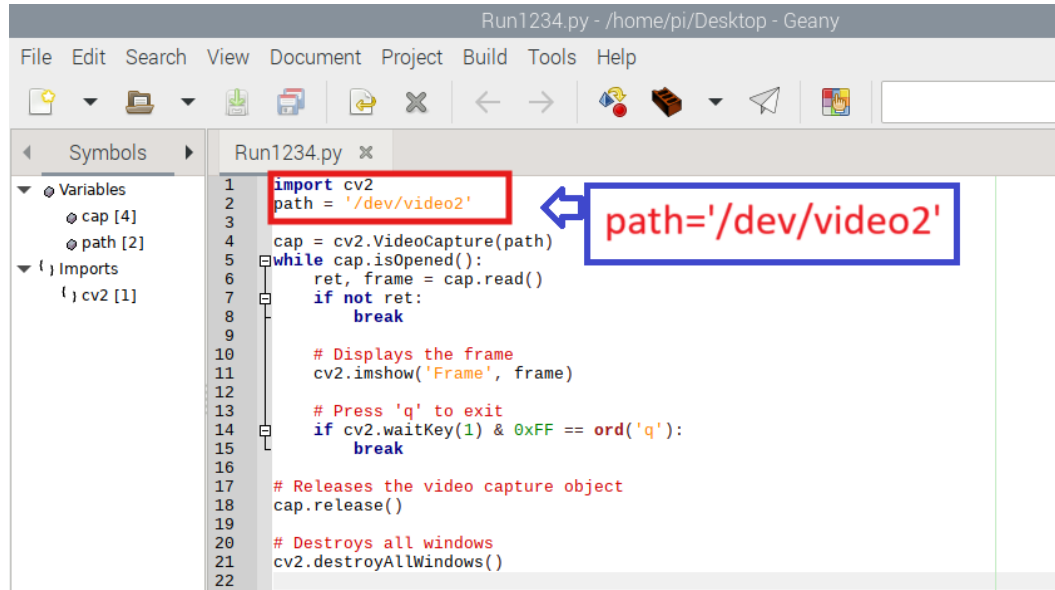
pi@iriv:~$ ffplay /dev/video2
ffplay version 4.3.8-0+deb11u3+rpt1 Copyright (c) 2000-2016 the FFmpeg developers
  built with gcc 10 (Debian 10.2.1-6)
  configuration: --prefix=/usr --extra-version=0
  libavutil      56. 51.100 / 56. 51.100
  libavcodec     58. 91.100 / 58. 91.100
  libavformat    58. 45.100 / 58. 45.100
  libavdevice    58. 10.100 / 58. 10.100
  libavfilter    7. 85.100 / 7. 85.100
  libavresample  4.  0.  0 / 4.  0.  0
  libswscale     5.  7.100 / 5.  7.100
  libswresample  3.  7.100 / 3.  7.100
  libpostproc   55.  7.100 / 55.  7.100
Input #0, video4linux2,v4l2, from '/dev/video2':
  Duration: N/A, start: 922.772293, bitrate: 147456 kb/s
  Stream #0:0: Video: rawvideo (YUY2 / 0x32595559), yuyv422, 640x480, 147456 kb/s, 30 fps, 30 tbr, 1000k tbn, 1000k tbc
  Metadata:
    b67.93 M-V: 1.085 fd= 389 aq= 0KB vq= 3600KB sq= 0B f=0/0
```



6.3.2.2b Open USB Camera – with vlc



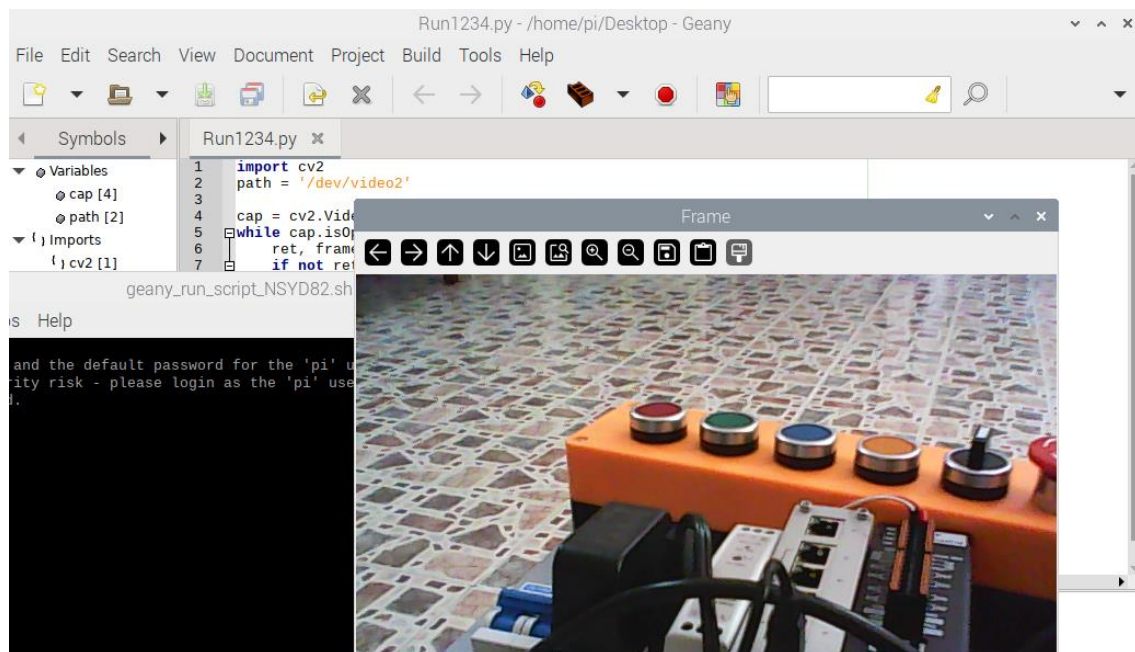
6.3.2.2c Open USB Camera – with python Code



```

1 import cv2
2 path = '/dev/video2'
3
4 cap = cv2.VideoCapture(path)
5 while cap.isOpened():
6     ret, frame = cap.read()
7     if not ret:
8         break
9
10    # Displays the frame
11    cv2.imshow('Frame', frame)
12
13    # Press 'q' to exit
14    if cv2.waitKey(1) & 0xFF == ord('q'):
15        break
16
17    # Releases the video capture object
18    cap.release()
19
20    # Destroys all windows
21    cv2.destroyAllWindows()
22

```



6.3.2.3a Open IP Camera – with vlc

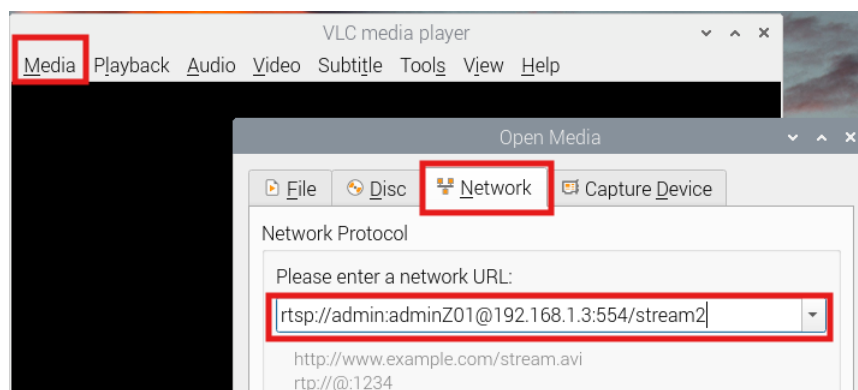
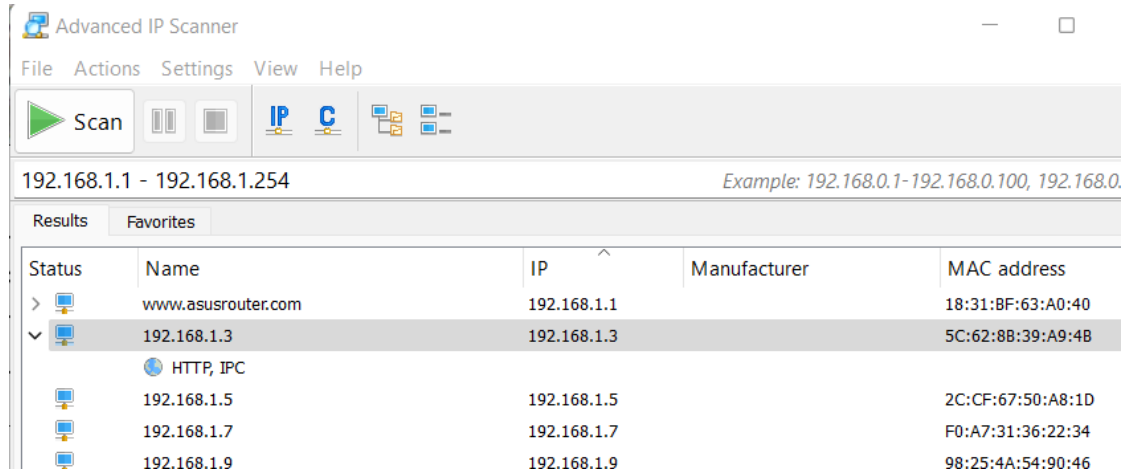
กรณี กล้องอยู่ในวงแลนเดียวกัน >> login=admin, Password=adminZ01, IP=192.168.1.3

TP-Link 1920x1080 >> rtsp://admin:adminZ01@192.168.1.3:554/stream1

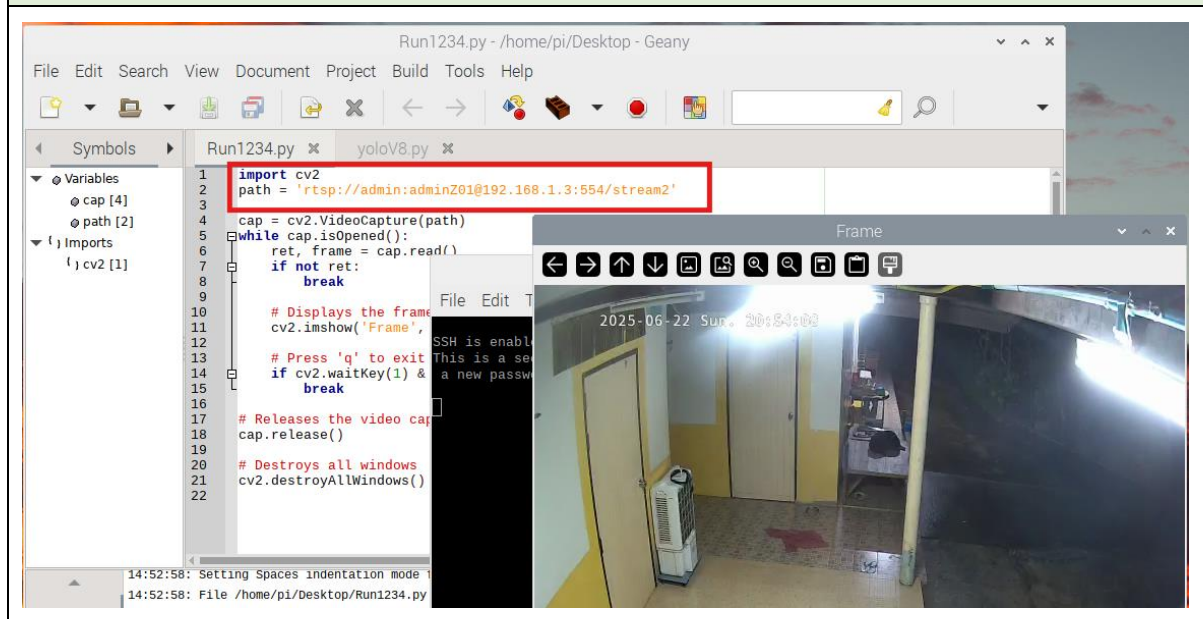
TP-Link 640x360 >> rtsp://admin:adminZ01@192.168.1.3:554/stream2

Hikvision Main Stream >> rtsp://admin:adminZ01@192.168.1.3:554/Streaming/Channels/101

Hikvision Sub Stream >> rtsp://admin:adminZ01@192.168.1.3:554/Streaming/Channels/102

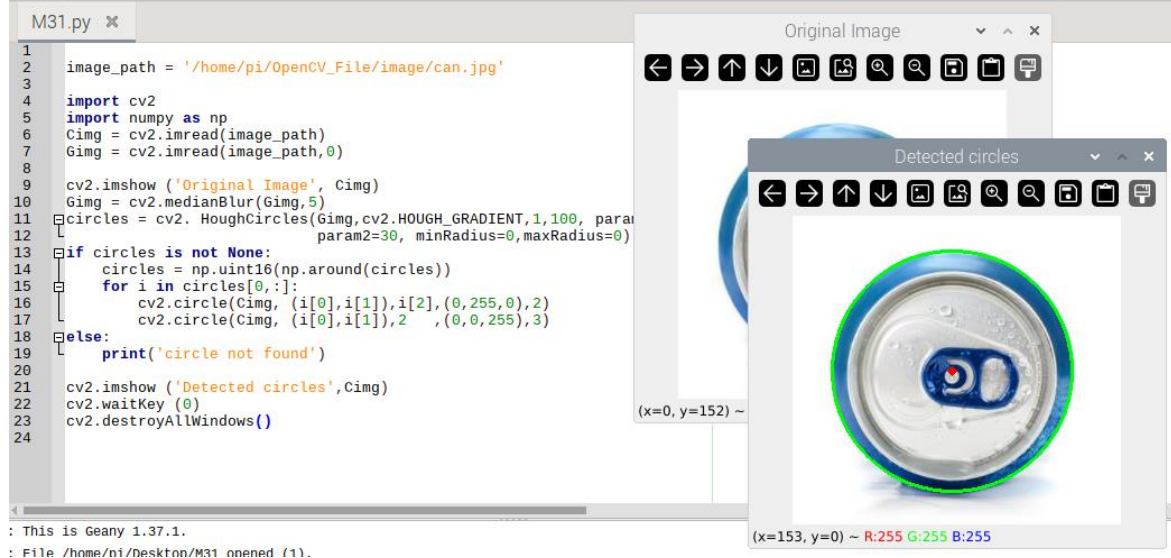


6.3.2.3b Open IP Camera – with python Code



6.3.3 -- Circle Detect

6.3.3.1 Hough Circle Example



```
image_path = '/home/pi/OpenCV_File/image/can.jpg'
```

```
import cv2
import numpy as np
Cimg = cv2.imread(image_path)
Gimg = cv2.imread(image_path, 0)
```

```
cv2.imshow('Original Image', Cimg)
Gimg = cv2.medianBlur(Gimg, 5)
circles = cv2.HoughCircles(Gimg, cv2.HOUGH_GRADIENT, 1, 100, param1=100, param2=30, minRadius=0, maxRadius=0)
if circles is not None:
    circles = np.uint16(np.round(circles))
    for i in circles[0,:]:
        cv2.circle(Cimg, (i[0],i[1]),i[2],(0,255,0),2)
        cv2.circle(Cimg, (i[0],i[1]),2,(0,0,255),3)
else:
    print('circle not found')
```

```
cv2.imshow('Detected circles', Cimg)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

cv2.HoughCircles(image, method, dp, minDist, param1, param2, minRadius, maxRadius)

- cv2.HOUGH_GRADIENT, # Method: Hough Gradient Method (วิธีเดียวที่ OpenCV รองรับในปัจจุบัน)
- 1 # dp: Inverse ratio of the accumulator resolution to the image resolution.
ถ้า dp=1, accumulator มี resolution เท่ากับภาพต้นฉบับ
ถ้า dp=2, accumulator มี resolution ครึ่งหนึ่ง
ค่านี้มักจะใช้ 1 หรือ 1.x
- 100 # minDist: Minimum distance between the centers of the detected circles.
ถ้าค่าน้อยไป อาจเจอวงกลมซ้อนกันหลายวง ถ้ามากไป อาจพลาดวงกลมบางวง
ในที่นี้ใช้ความสูงของภาพหาร 8 เป็นค่าเริ่มต้น
- param1=100, # param1: Upper threshold for the Canny edge detector (lower is half of this).
ค่าที่สูงขึ้นจะทำให้ Canny ตรวจจับขอบที่แรงขึ้นเท่านั้น (ลด false positives)
- param2=30, # param2: Accumulator threshold for the circle centers at the detection stage.
ค่าที่น้อยลงจะตรวจจับวงกลมได้มากขึ้น (รวมถึง false positives)
ค่าที่มากขึ้นจะกรองวงกลมที่ไม่ชัดเจนออกไป
- minRadius=0 # minRadius: Minimum circle radius in pixels.
- maxRadius=0 # maxRadius: Maximum circle radius in pixels. (ถ้าเป็น 0 หมายถึงไม่จำกัดสูงสุด)

6.3.3.2 Coin Count

```

M31_HoughCircle.py x M32_Coin.py x
1
2 image_path = '/home/pi/OpenCV_File/image/coins1.jpg'
3
4 import cv2
5 import numpy as np
6 Cimg = cv2.imread(image_path)
7 Gimg = cv2.imread(image_path,0)
8
9 cv2.imshow ('Original Image', Cimg)
10 imgBlur = cv2.medianBlur(Gimg,7)
11 imgEdges = cv2.Canny (imgBlur, 50, 100, apertureSize=5, L2gradient=True)
12 circles = cv2. HoughCircles(imgEdges,cv2.HOUGH_GRADIENT,1,50, param1=50,
13                             param2=20, minRadius=30,maxRadius=50)
14 if circles is not None:
15     circles = np.uint16(np.around(circles))
16     for i in circles[0,:]:
17         cv2.circle(Cimg, (i[0],i[1]),i[2],(0,255,0),2)
18         cv2.circle(Cimg, (i[0],i[1]),2 ,(0,0,255),3)
19 else:
20     print('circle not found')
21
22 cv2.imshow ('Detected circles',Cimg)
23 cv2.waitKey (0)
24 cv2.destroyAllWindows()
25

```

```

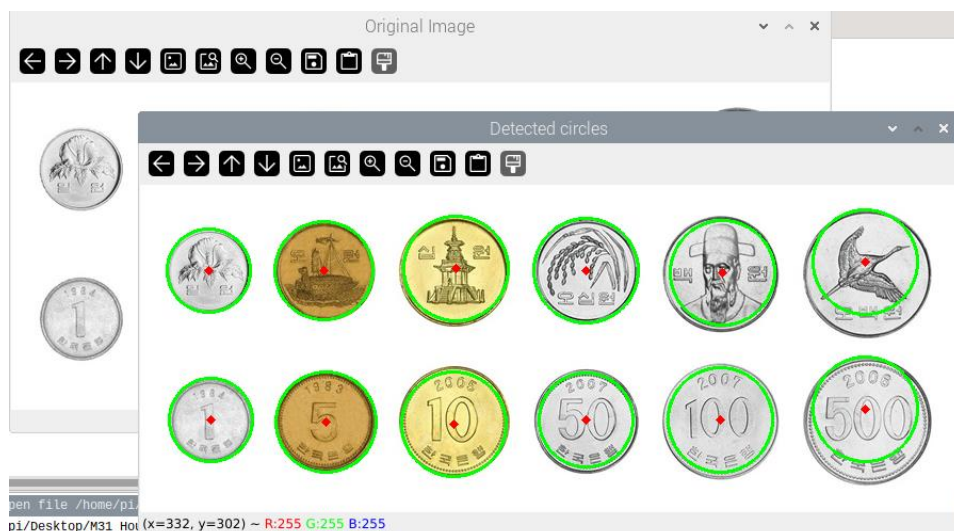
image_path = '/home/pi/OpenCV_File/image/coins1.jpg'

import cv2
import numpy as np
Cimg = cv2.imread(image_path)
Gimg = cv2.imread(image_path,0)

cv2.imshow ('Original Image', Cimg)
imgBlur = cv2.medianBlur(Gimg,7)
imgEdges = cv2.Canny (imgBlur, 50, 100, apertureSize=5, L2gradient=True)
circles = cv2. HoughCircles(imgEdges,cv2.HOUGH_GRADIENT,1,50, param1=50, param2=20, minRadius=30, maxRadius=50)
if circles is not None:
    circles = np.uint16(np.around(circles))
    for i in circles[0,:]:
        cv2.circle(Cimg, (i[0],i[1]),i[2],(0,255,0),2)
        cv2.circle(Cimg, (i[0],i[1]),2 ,(0,0,255),3)
else:
    print('circle not found')

cv2.imshow ('Detected circles',Cimg)
cv2.waitKey (0)
cv2.destroyAllWindows()

```

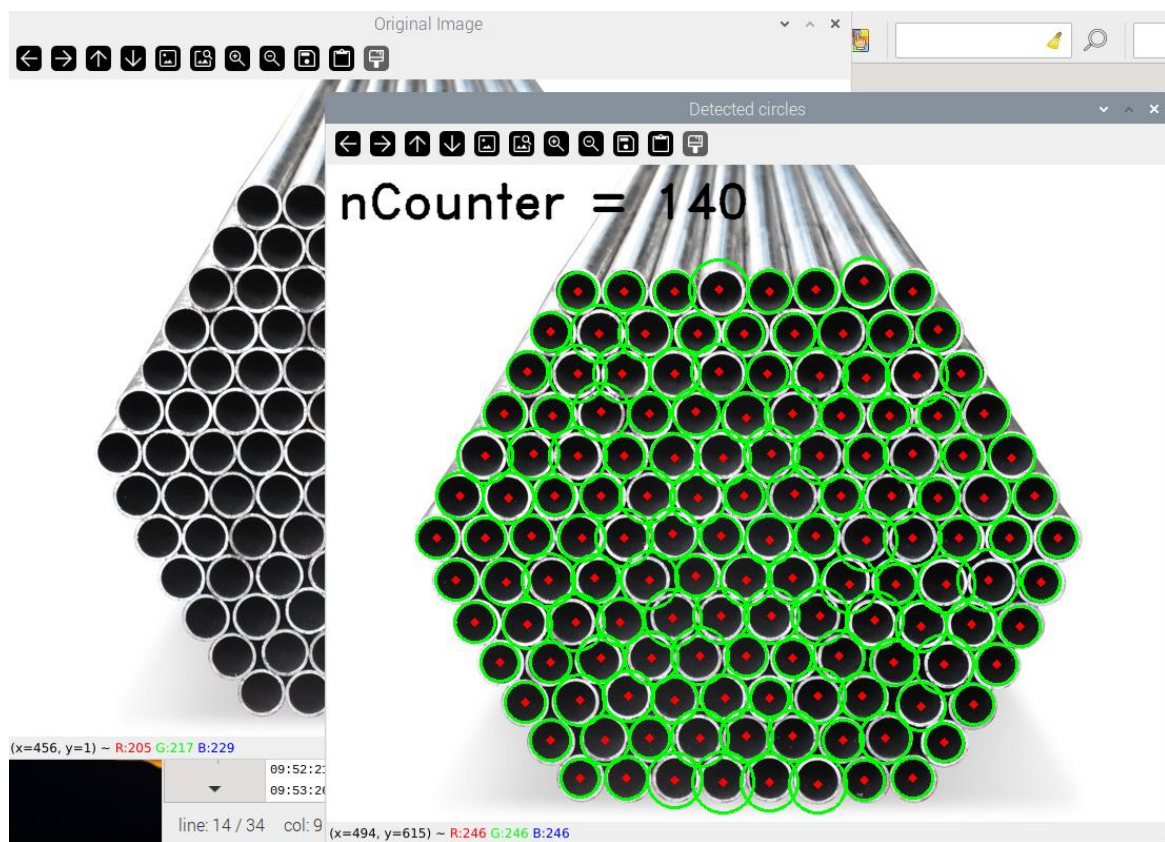


6.3.3.3 Pipe Count

```

M33_Pipe.py x
1
2 image_path = '/home/pi/OpenCV_File/image/XX.jpg'
3
4 import cv2
5 import numpy as np
6 Cimg = cv2.imread(image_path)
7 Gimg = cv2.imread(image_path,0)
8
9 cv2.imshow ('Original Image', Cimg)
10 imgBlur = cv2.medianBlur(Gimg,7)
11 imgEdges = cv2.Canny (imgBlur, 50, 100, apertureSize=5, L2gradient=True)
12 circles = cv2. HoughCircles(imgEdges,cv2.HOUGH_GRADIENT,1,35, param1=25,
13                             param2=30, minRadius=15, maxRadius=30)
14 Count = 0
15 if circles is not None:
16     circles = np.uint16(np.around(circles))
17     for i in circles[0,:]:
18         cv2.circle(Cimg, (i[0],i[1]),i[2],(0,255,0),2)
19         cv2.circle(Cimg, (i[0],i[1]),2,(0,0,255),3)
20         Count = Count + 1
21 else:
22     print('circle not found')
23
24 TEXT_FACE = cv2.FONT_HERSHEY_DUPLEX
25 TEXT_SCALE = 1.5
26 TEXT_THICKNESS = 2
27 Text = 'nCounter = ' + str(Count)
28 cv2.putText(Cimg, Text, (10,50), TEXT_FACE, TEXT_SCALE, (0,0,0), TEXT_THICKNESS, cv2.LINE_AA)
29 print(Text)
30
31 cv2.imshow ('Detected circles',Cimg)
32 cv2.waitKey (0)
33 cv2.destroyAllWindows()
34

```



```
image_path = '/home/pi/OpenCV_File/image/XX.jpg'

import cv2
import numpy as np
Cimg = cv2.imread(image_path)
Gimg = cv2.imread(image_path,0)

cv2.imshow ('Original Image', Cimg)
imgBlur = cv2.medianBlur(Gimg,7)
imgEdges = cv2.Canny (imgBlur, 50, 100, apertureSize=5, L2gradient=True)
circles = cv2. HoughCircles(imgEdges, cv2.HOUGH_GRADIENT,1,35, param1=25, param2=30, minRadius=15, maxRadius=30)
Count = 0
if circles is not None:
    circles = np.uint16(np.around(circles))
    for i in circles[0,:]:
        cv2.circle(Cimg, (i[0],i[1]),i[2],(0,255,0),2)
        cv2.circle(Cimg, (i[0],i[1]),2 ,(0,0,255),3)
        Count = Count + 1
    else:
        print('circle not found')

TEXT_FACE = cv2.FONT_HERSHEY_DUPLEX
TEXT_SCALE = 1.5
TEXT_THICKNESS = 2
Text = 'nCounter = ' + str(Count)
cv2.putText(Cimg, Text, (10,50), TEXT_FACE, TEXT_SCALE, (0,0,0), TEXT_THICKNESS, cv2.LINE_AA)
print(Text)

cv2.imshow ('Detected circles',Cimg)
cv2.waitKey (0)
cv2.destroyAllWindows()
```

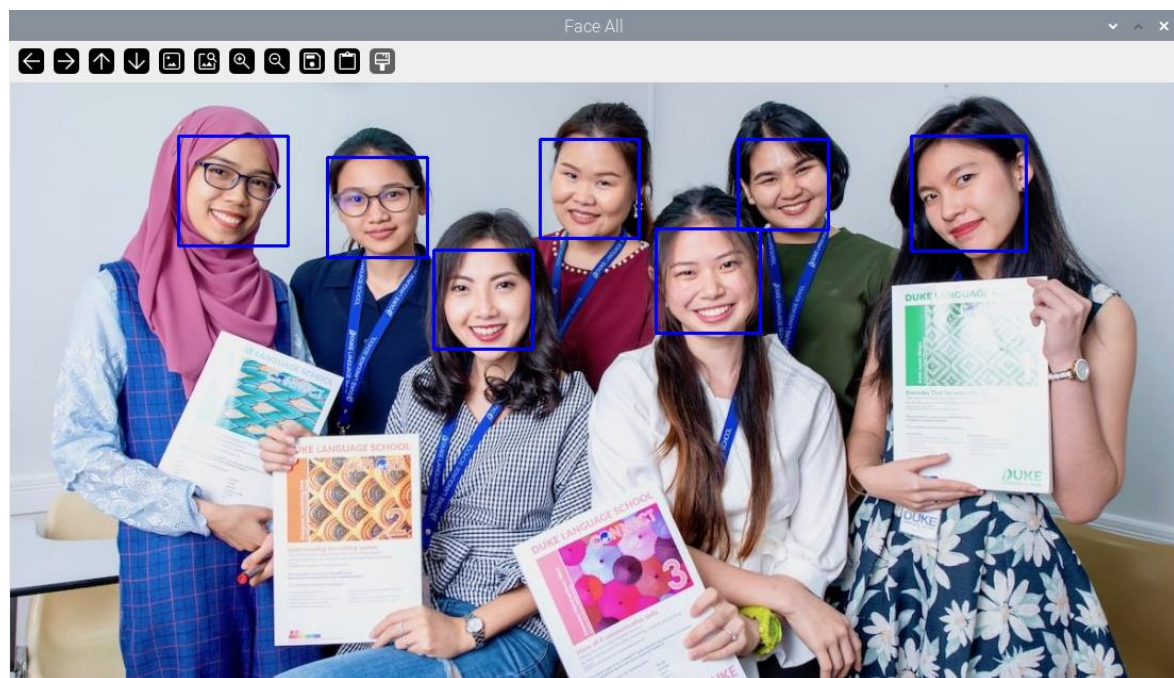
6.3.4 -- Face Detection, Gender and Age Detection

6.3.4.1 Face Detection with Haar Cascade Model

```

41_Face.py x 42_FaceEye.py x 43_GenAge.py x
1
2 import cv2
3 face_cascade = cv2.CascadeClassifier('/home/pi/OpenCV_File/data/haarcascade_frontalface_default.xml')
4 imageC = cv2.imread('/home/pi/OpenCV_File/image/Face101.jpg')
5 #imageC = cv2.imread('/home/pi/OpenCV_File/image/Face102.jpg')
6
7 imageG = cv2.cvtColor(imageC, cv2.COLOR_BGR2GRAY)
8 faces = face_cascade.detectMultiScale(imageG, 1.1, 4)
9 Count = 1
10 print("Person = ", len(faces))
11 for (x, y, w, h) in faces:
12     cv2.rectangle(imageC, (x, y), (x + w, y + h), (255, 0, 0), 2)
13     ROI_imageG = imageG[y:y+h, x:x+w]
14     ROI_imageC = imageC[y:y+h, x:x+w]
15     Count = Count + 1
16     cv2.imshow("Face Single" + str(Count), ROI_imageC)
17
18 cv2.imshow("Face All", imageC)
19 cv2.waitKey(0)
20 cv2.destroyAllWindows()
21

```



```

import cv2
face_cascade = cv2.CascadeClassifier('/home/pi/OpenCV_File/data/haarcascade_frontalface_default.xml')
imageC = cv2.imread('/home/pi/OpenCV_File/image/Face101.jpg')
#imageC = cv2.imread('/home/pi/OpenCV_File/image/Face102.jpg')

imageG = cv2.cvtColor(imageC, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(imageG, 1.1, 4)
Count = 1
print("Person = ", len(faces))
for (x, y, w, h) in faces:
    cv2.rectangle(imageC, (x, y), (x + w, y + h), (255, 0, 0), 2)
    ROI_imageG = imageG[y:y+h, x:x+w]
    ROI_imageC = imageC[y:y+h, x:x+w]
    Count = Count + 1
    cv2.imshow("Face Single" + str(Count), ROI_imageC)

cv2.imshow("Face All", imageC)
cv2.waitKey(0)
cv2.destroyAllWindows()

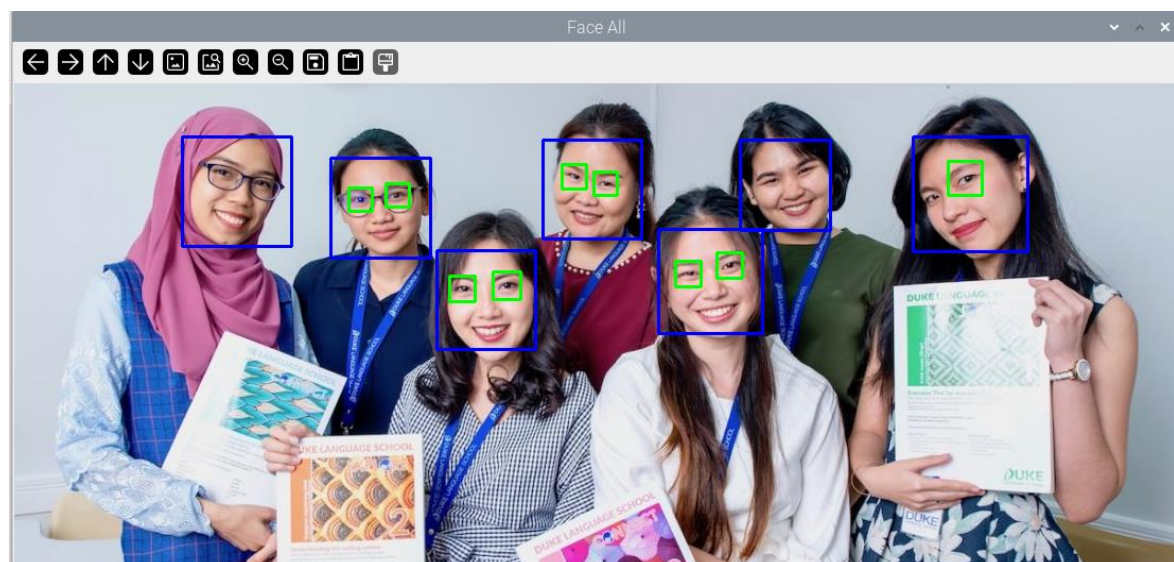
```

6.3.4.2 Face and Eye Detection with Haar Cascade Model

```

42_FaceEye.py x 43_GenAge.py x 41_Face.py x
1
2 import cv2
3 face_cascade = cv2.CascadeClassifier('/home/pi/OpenCV_File/data/haarcascade_frontalface_default.xml')
4 eye_cascade = cv2.CascadeClassifier('/home/pi/OpenCV_File/data/haarcascade_eye.xml')
5 imageC = cv2.imread('/home/pi/OpenCV_File/image/Face101.jpg')
6 #imageC = cv2.imread('/home/pi/OpenCV_File/image/Fac102.jpg')
7
8 imageG = cv2.cvtColor(imageC, cv2.COLOR_BGR2GRAY)
9 faces = face_cascade.detectMultiScale(imageG, 1.1, 4)
10 print("Person = ", len(faces))
11 for (x, y, w, h) in faces:
12     cv2.rectangle(imageC, (x, y), (x + w, y + h), (255, 0, 0), 2)
13     ROI_imageG = imageG[y:y+h, x:x+w]
14     ROI_imageC = imageC[y:y+h, x:x+w]
15     eye = eye_cascade.detectMultiScale(ROI_imageG)
16     for (ex, ey, ew, eh) in eye:
17         cv2.rectangle(ROI_imageC, (ex, ey), (ex + ew, ey + eh), (0, 255, 0), 2)
18
19 cv2.imshow("Face All", imageC)
20 cv2.waitKey(0)
21 cv2.destroyAllWindows()
22

```



```

import cv2
face_cascade = cv2.CascadeClassifier('/home/pi/OpenCV_File/data/haarcascade_frontalface_default.xml')
eye_cascade = cv2.CascadeClassifier('/home/pi/OpenCV_File/data/haarcascade_eye.xml')
imageC = cv2.imread('/home/pi/OpenCV_File/image/Face101.jpg')
#imageC = cv2.imread('/home/pi/OpenCV_File/image/Fac102.jpg')

imageG = cv2.cvtColor(imageC, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(imageG, 1.1, 4)
print("Person = ", len(faces))
for (x, y, w, h) in faces:
    cv2.rectangle(imageC, (x, y), (x + w, y + h), (255, 0, 0), 2)
    ROI_imageG = imageG[y:y+h, x:x+w]
    ROI_imageC = imageC[y:y+h, x:x+w]
    eye = eye_cascade.detectMultiScale(ROI_imageG)
    for (ex, ey, ew, eh) in eye:
        cv2.rectangle(ROI_imageC, (ex, ey), (ex + ew, ey + eh), (0, 255, 0), 2)

cv2.imshow("Face All", imageC)
cv2.waitKey(0)
cv2.destroyAllWindows()

```

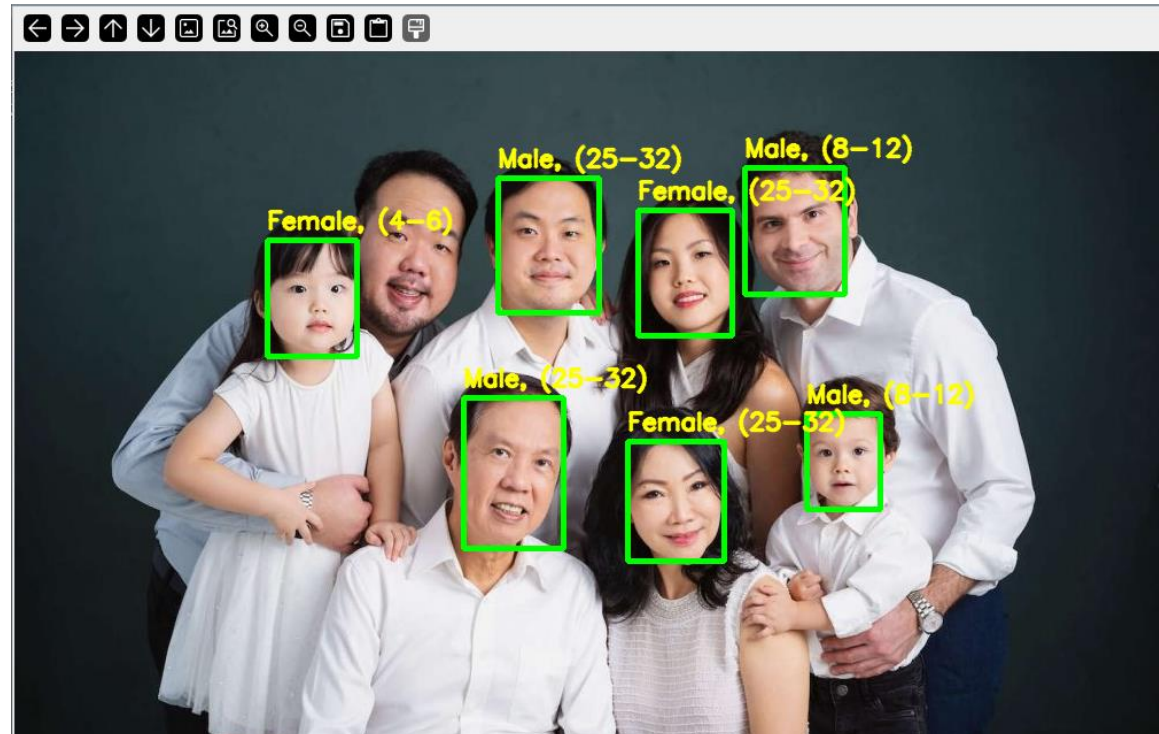
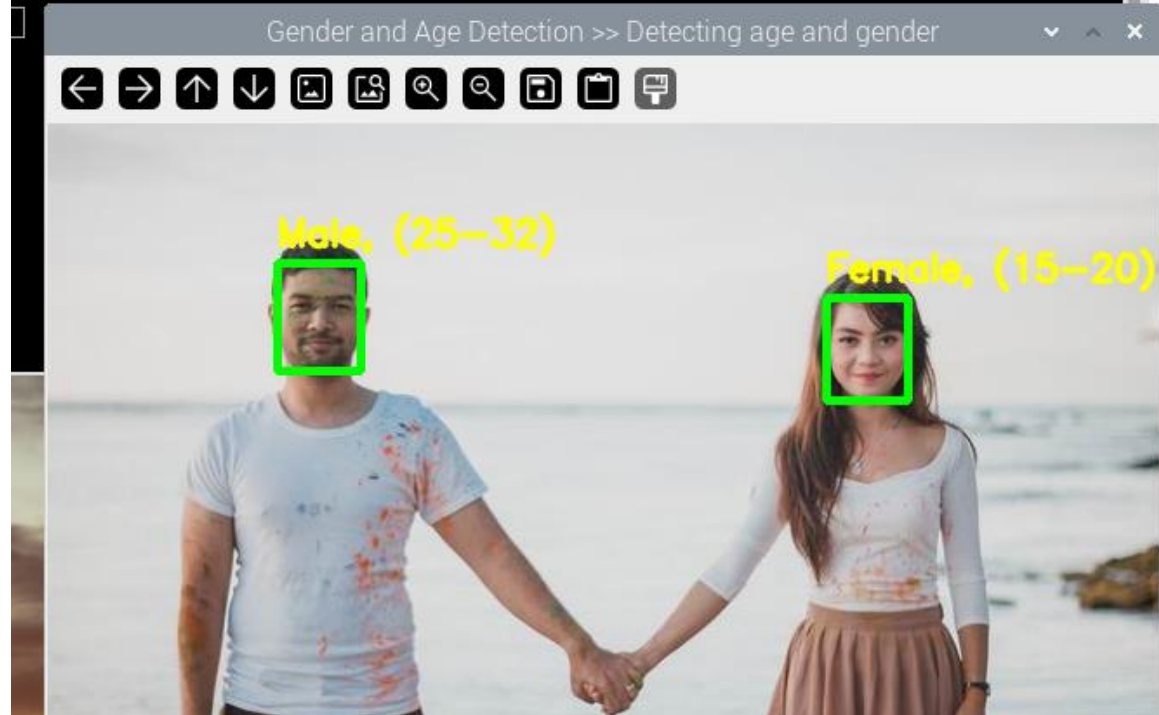
6.3.4.3 Face Detection with Haar Cascade Model

```

43_GenAge.py x
1
2 import cv2
3 import math
4 import argparse
5
6 myName = 'Gender and Age Detection'
7 imageC = cv2.imread("/home/pi/OpenCV_File/image/GenAge/test01.jpg")
8
9 def highlightFace(net, frame, conf_threshold=0.7):
10     frameOpencvDnn=frame.copy()
11     frameHeight=frameOpencvDnn.shape[0]
12     frameWidth=frameOpencvDnn.shape[1]
13     blob=cv2.dnn.blobFromImage(frameOpencvDnn, 1.0, (300, 300), [104, 117, 123], True, False)
14
15     net.setInput(blob)
16     detections=net.forward()
17     faceBoxes=[]
18     for i in range(detections.shape[2]):
19         confidence=detections[0,0,i,2]
20         if confidence>conf_threshold:
21             x1=int(detections[0,0,i,3]*frameWidth)
22             y1=int(detections[0,0,i,4]*frameHeight)
23             x2=int(detections[0,0,i,5]*frameWidth)
24             y2=int(detections[0,0,i,6]*frameHeight)
25             faceBoxes.append([x1,y1,x2,y2])
26             cv2.rectangle(frameOpencvDnn, (x1,y1), (x2,y2), (0,255,0), int(round(frameHeight/150)), 8)
27     return frameOpencvDnn,faceBoxes
28
29
30 faceProto = "/home/pi/OpenCV_File/data/opencv_face_detector.pbtxt"
31 faceModel = "/home/pi/OpenCV_File/data/opencv_face_detector_uint8.pb"
32 ageProto = "/home/pi/OpenCV_File/data/age_deploy.prototxt"
33 ageModel = "/home/pi/OpenCV_File/data/age_net.caffemodel"
34 genderProto = "/home/pi/OpenCV_File/data/gender_deploy.prototxt"
35 genderModel = "/home/pi/OpenCV_File/data/gender_net.caffemodel"
36
37 MODEL_MEAN_VALUES = (78.4263377603, 87.7689143744, 114.895847746)
38 ageList = ['(0-2)', '(4-6)', '(8-12)', '(15-20)', '(25-32)', '(38-43)', '(48-53)', '(60-100)']
39 genderList = ['Male', 'Female']
40
41 faceNet = cv2.dnn.readNet(faceModel, faceProto)
42 ageNet = cv2.dnn.readNet(ageModel, ageProto)
43 genderNet = cv2.dnn.readNet(genderModel, genderProto)
44
45 padding=20
46 frame = imageC.copy()
47
48 resultImg, faceBoxes=highlightFace(faceNet, frame)
49 if not faceBoxes:
50     print("No face detected")
51
52 for faceBox in faceBoxes:
53     face=frame[max(0,faceBox[1]-padding):min(faceBox[3]+padding, frame.shape[0]-1),
54               max(0,faceBox[0]-padding):min(faceBox[2]+padding, frame.shape[1]-1)]
55
56     blob=cv2.dnn.blobFromImage(face, 1.0, (227,227), MODEL_MEAN_VALUES, swapRB=False)
57     genderNet.setInput(blob)
58     genderPreds=genderNet.forward()
59     gender=genderList[genderPreds[0].argmax()]
60     print(f'Gender: {gender}')
61
62     ageNet.setInput(blob)
63     agePreds=ageNet.forward()
64     age=ageList[agePreds[0].argmax()]
65     print(f'Age: {age[1:-1]} years')
66     print()
67
68     cv2.putText(resultImg, f'{gender}, {age}', (faceBox[0], faceBox[1]-10),
69                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,255), 2, cv2.LINE_AA)
70     cv2.imshow(myName + " >> Detecting age and gender", resultImg)
71
72 cv2.waitKey(0)
73 cv2.destroyAllWindows()
74

```

Gender: Female
Age: 15-20 years
Gender: Male
Age: 25-32 years



```

import cv2
import math
import argparse

myName = 'Gender and Age Detection'
imageC = cv2.imread("/home/pi/OpenCV_File/image/GenAge/test01.jpg")

def highlightFace(net, frame, conf_threshold=0.7):
    frameOpencvDnn=frame.copy()
    frameHeight=frameOpencvDnn.shape[0]
    frameWidth=frameOpencvDnn.shape[1]
    blob=cv2.dnn.blobFromImage(frameOpencvDnn, 1.0, (300, 300), [104, 117, 123], True, False)

    net.setInput(blob)
    detections=net.forward()
    faceBoxes=[]
    for i in range(detections.shape[2]):
        confidence=detections[0,0,i,2]
        if confidence>conf_threshold:
            x1=int(detections[0,0,i,3]*frameWidth)
            y1=int(detections[0,0,i,4]*frameHeight)
            x2=int(detections[0,0,i,5]*frameWidth)
            y2=int(detections[0,0,i,6]*frameHeight)
            faceBoxes.append([x1,y1,x2,y2])
            cv2.rectangle(frameOpencvDnn, (x1,y1), (x2,y2), (0,255,0), int(round(frameHeight/150)), 8)
    return frameOpencvDnn,faceBoxes

faceProto = "/home/pi/OpenCV_File/data/opencv_face_detector.pbtxt"
faceModel = "/home/pi/OpenCV_File/data/opencv_face_detector_uint8.pb"
ageProto = "/home/pi/OpenCV_File/data/age_deploy.prototxt"
ageModel = "/home/pi/OpenCV_File/data/age_net.caffemodel"
genderProto = "/home/pi/OpenCV_File/data/gender_deploy.prototxt"
genderModel = "/home/pi/OpenCV_File/data/gender_net.caffemodel"

MODEL_MEAN_VALUES = (78.4263377603, 87.7689143744, 114.895847746)
ageList = ['(0-2)', '(4-6)', '(8-12)', '(15-20)', '(25-32)', '(38-43)', '(48-53)', '(60-100)']
genderList = ['Male', 'Female']

faceNet = cv2.dnn.readNet(faceModel,faceProto)
ageNet = cv2.dnn.readNet(ageModel,ageProto)
genderNet = cv2.dnn.readNet(genderModel,genderProto)

padding=20
frame = imageC.copy()

resultImg,faceBoxes=highlightFace(faceNet,frame)
if not faceBoxes:
    print("No face detected")

for faceBox in faceBoxes:
    face=frame[max(0,faceBox[1]-padding):min(faceBox[3]+padding, frame.shape[0]-1),
               max(0,faceBox[0]-padding):min(faceBox[2]+padding, frame.shape[1]-1)]

    blob=cv2.dnn.blobFromImage(face, 1.0, (227,227), MODEL_MEAN_VALUES, swapRB=False)
    genderNet.setInput(blob)
    genderPreds=genderNet.forward()
    gender=genderList[genderPreds[0].argmax()]
    print(f'Gender: {gender}')

    ageNet.setInput(blob)
    agePreds=ageNet.forward()
    age=ageList[agePreds[0].argmax()]
    print(f'Age: {age[1:-1]} years')
    print()

    cv2.putText(resultImg, f'{gender}, {age}', (faceBox[0], faceBox[1]-10),
                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,255), 2, cv2.LINE_AA)
    cv2.imshow(myName + " >> Detecting age and gender", resultImg)

cv2.waitKey(0)
cv2.destroyAllWindows()

```

6.3.5 -- Yolo8

ติดตั้ง Ultralytics

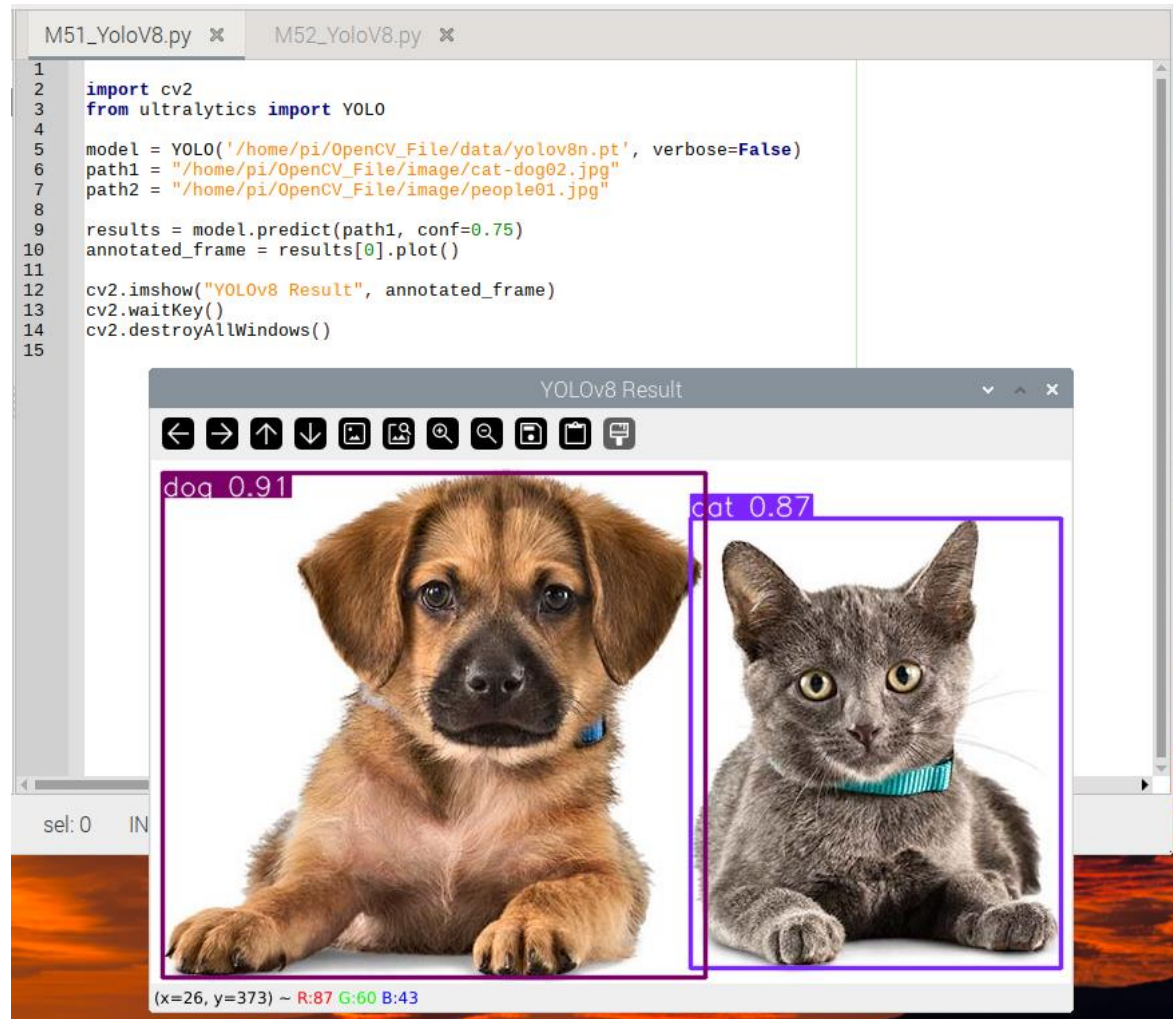
pip install ultralytics

```

File Edit Tabs Help
pi@iriv:~ $ pip install ultralytics
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting ultralytics
  Downloading ultralytics-8.3.155-py3-none-any.whl (1.0 MB)
    |#####| 1.0 MB 6.1 MB/s
Requirement already satisfied: psutil in /usr/lib/python3/dist-packages (from ultralytics) (5.8.0)
Collecting pyyaml>=5.3.1
  Downloading PyYAML-6.0.2-cp39-cp39-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (720 kB)
    |#####| 720 kB 11.7 MB/s
Requirement already satisfied: requests>=2.23.0 in /usr/lib/python3/dist-packages (from ultralytics) (2.25.1)
Requirement already satisfied: pillow>=7.1.2 in /usr/lib/python3/dist-packages (from ultralytics) (8.1.2)
Collecting tqdm>=4.64.0
  Downloading https://www.piwheels.org/simple/tqdm/tqdm-4.67.1-py3-none-any.whl (78 kB)
    |#####| 78 kB 221 kB/s
Collecting numpy>=1.23.0
  Downloading numpy-2.0.2-cp39-cp39-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (13.9 MB)
    |#####| 13.9 MB 11.2 MB/s
Collecting ultralytics-thop>=2.0.0
  Downloading https://www.piwheels.org/simple/ultralytics-thop/ultralytics_thop-2.0.14-py3-none-any.whl (26 kB)
Collecting scipy>=1.4.1
  Downloading scipy-1.13.1-cp39-cp39-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (33.7 MB)
    |#####| 33.7 MB 15.6 MB/s
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
WARNING: The script cpuinfo is installed in '/home/pi/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
WARNING: The scripts ultralytics and yolo are installed in '/home/pi/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed contourpy-1.3.0 cycycler-0.12.1 filelock-3.18.0 fonttools-4.58.4 fsspec-2025.5.1 importlib-resources-6.4.0
matplotlib-3.9.4 mpmath-1.3.0 networkx-3.2.1 numpy-2.0.2 packaging-25.0 pandas-2.3.0 py-cpuinfo-9.0.0 pyparsing-3.2.3 python-3.9.20
pytz-2025.2 pyyaml-6.0.2 scipy-1.13.1 sympy-1.14.0 torch-2.7.1 torchvision-0.22.1 tqdm-4.67.1 typing-extensions-4.14.0 tz
ics-8.3.155 ultralytics-thop-2.0.14 zipp-3.23.0
pi@iriv:~ $
pi@iriv:~ $

```

6.3.5.1 -- yoloV8 Image Test



```

import cv2
from ultralytics import YOLO

model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt', verbose=False)
path1 = "/home/pi/OpenCV_File/image/cat-dog02.jpg"
path2 = "/home/pi/OpenCV_File/image/people01.jpg"

results = model.predict(path1, conf=0.75)
annotated_frame = results[0].plot()

cv2.imshow("YOLOv8 Result", annotated_frame)
cv2.waitKey()
cv2.destroyAllWindows()

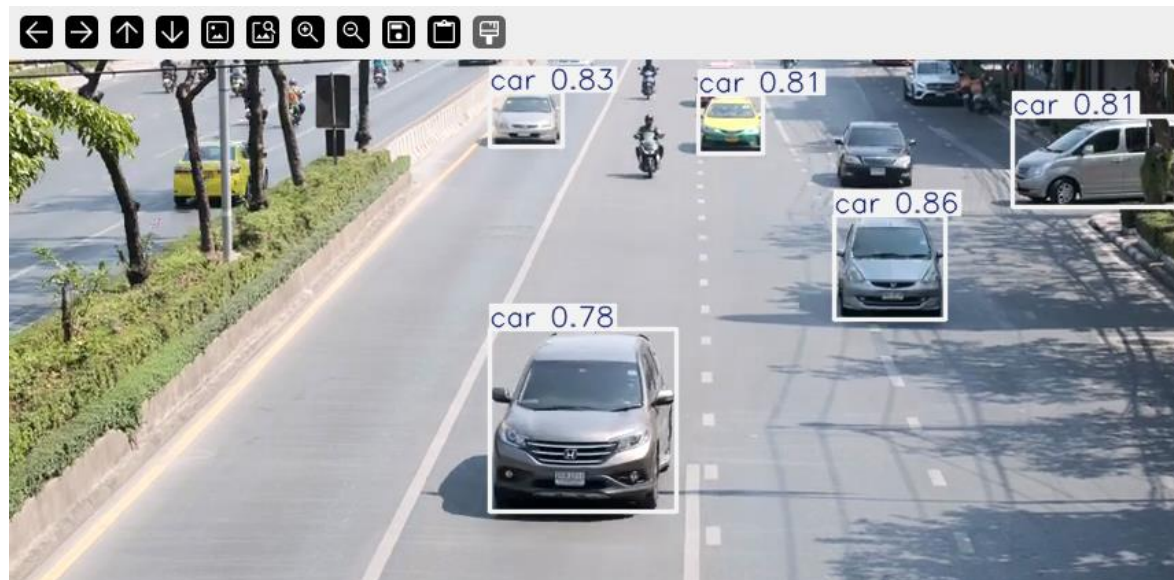
```

6.3.5.2 -- yoloV8 Video Test

```

M51_YoloV8.py x M52_YoloV8.py x
1 import cv2
2 from ultralytics import YOLO
3
4 model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt', verbose=False)
5 video_path = "/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4"
6
7 cap = cv2.VideoCapture(video_path)
8 while cap.isOpened():
9     success, frame = cap.read()
10    if success:
11        results = model(frame, conf=0.75, verbose=True)
12        annotated_frame = results[0].plot()
13        cv2.imshow("YOLOv8 Result", annotated_frame)
14        if cv2.waitKey(1) & 0xFF == ord("q"):
15            break
16    else:
17        break
18
19 cap.release()
20 cv2.destroyAllWindows()
21

```



```

import cv2
from ultralytics import YOLO

model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt', verbose=False)
video_path = "/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4"

cap = cv2.VideoCapture(video_path)
while cap.isOpened():
    success, frame = cap.read()
    if success:
        results = model(frame, conf=0.75, verbose=True)
        annotated_frame = results[0].plot()
        cv2.imshow("YOLOv8 Result", annotated_frame)
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        break

cap.release()
cv2.destroyAllWindows()

```

6.3.6 -- Application1/2: Contour Counter

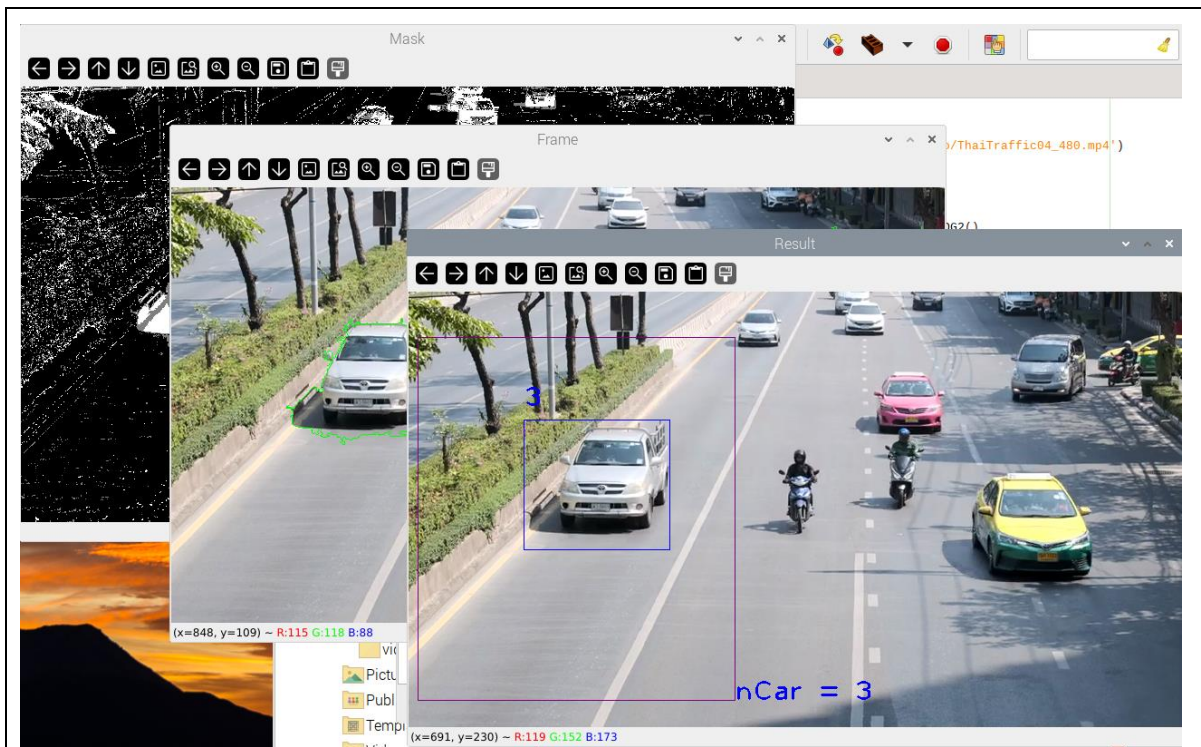
```

M61_ContourCount.py x
1
2 # 10-with Tracker
3 import cv2
4 cap = cv2.VideoCapture('/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4')
5 roiX,roiY,roiW,roiH = 10, 50, 350, 400
6
7 from M60_tracker import *
8 tracker = EuclideanDistTracker()
9
10 object_detector = cv2.createBackgroundSubtractorMOG2()
11 font = cv2.FONT_HERSHEY_PLAIN
12 nCar = 0
13
14 while(cap.isOpened()):
15     ret, frame = cap.read()
16     if frame is None:
17         break
18
19     result = frame.copy()
20     mask = object_detector.apply(frame)
21     contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
22     detections = []
23     for cnt in contours:
24         # Calculate area and remove small elements
25         area = cv2.contourArea(cnt)
26         if area > 4000:
27             cv2.drawContours(frame, [cnt], -1, (0, 255, 0), 1)
28             x, y, w, h = cv2.boundingRect(cnt)
29             if (x > roiX) and (y > roiY) and (roiX+roiW > x+w) and (roiY+roiH > y+h)):
30                 detections.append([x, y, w, h])
31
32     boxes_ids = tracker.update(detections)
33     for box_id in boxes_ids:
34         x, y, w, h, id = box_id
35         nCar = id
36         cv2.putText(result, str(id), (x, y - 15), font, 2, (255, 0, 0), 2)
37         cv2.rectangle(result, (x, y), (x+w, y+h), (255, 0, 0), 1)
38
39     cv2.rectangle(result, (roiX, roiY), (roiX+roiW, roiY+roiH), (125, 0, 125), 1)
40     cv2.putText(result, "nCar = " + str(nCar), (roiX+roiW, roiY+roiH), font, 2, (255, 0, 0), 2)
41
42     cv2.imshow("Mask", mask)
43     cv2.imshow("Frame", frame)
44     cv2.imshow("Result", result)
45     if cv2.waitKey(1) & 0xFF == ord("q"):
46         break
47     cap.release()
48     cv2.destroyAllWindows()
49

```

File “M60_tracker.py” กับ “M61_ContourCount.py” ต้องอยู่ Folder เดียวกัน





```
# 10-with Tracker
import cv2
cap = cv2.VideoCapture('/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4')
roiX,roiY,roiW,roiH = 10, 50, 350, 400

from M60_tracker import *
tracker = EuclideanDistTracker()

object_detector = cv2.createBackgroundSubtractorMOG2()
font = cv2.FONT_HERSHEY_PLAIN
nCar = 0

while(cap.isOpened()):
    ret, frame = cap.read()
    if frame is None:
        break

    result = frame.copy()
    mask = object_detector.apply(frame)
    contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
    detections = []
    for cnt in contours:
        # Calculate area and remove small elements
        area = cv2.contourArea(cnt)
        if area > 4000:
            cv2.drawContours(frame, [cnt], -1, (0, 255, 0), 1)
            x, y, w, h = cv2.boundingRect(cnt)
            if ((x > roiX) and (y > roiY) and (roiX+roiW > x+w) and (roiY+roiH > y+h)):
                detections.append([x, y, w, h])

    boxes_ids = tracker.update(detections)
    for box_id in boxes_ids:
        x, y, w, h, id = box_id
        nCar = id
        cv2.putText(result, str(id), (x, y - 15), font, 2, (255, 0, 0), 2)
        cv2.rectangle(result, (x, y), (x+w, y+h), (255, 0, 0), 1)

    cv2.rectangle(result, (roiX, roiY), (roiX+roiW, roiY+roiH), (125, 0, 125), 1)
    cv2.putText(result, "nCar = " + str(nCar), (roiX+roiW, roiY+roiH), font, 2, (255, 0, 0), 2)

    cv2.imshow("Mask", mask)
    cv2.imshow("Frame", frame)
    cv2.imshow("Result", result)
    if cv2.waitKey(1) & 0xFF == ord("q"):
        break
    cap.release()
cv2.destroyAllWindows()
```

6.3.7 -- Application2/2: Car Counter with Yolo

6.3.7.1 -- Run yoloV8

```

M71_yoloV8.py x M72_yoloV8_Count.py x
1
2 import cv2
3 from ultralytics import YOLO
4
5 model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt', verbose=False)
6 video_path = "/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4"
7
8 roiX,roiY,roiW,roiH = 10, 150, 280, 300
9 frame_kStep, frame_count = 8, 0
10
11 cap = cv2.VideoCapture(video_path)
12 while cap.isOpened():
13     frame_count += 1
14     success, mainFrame = cap.read()
15     if success:
16         if frame_count % frame_kStep == 0:
17             frame = mainFrame[roiY:roiY+roiH, roiX:roiX+roiW]
18             results = model(frame, conf=0.55, verbose=True)
19             annotated_frame = results[0].plot()
20             mainFrame[roiY:roiY+roiH, roiX:roiX+roiW] = annotated_frame
21             cv2.rectangle(mainFrame, (roiX,roiY), (roiX+roiW,roiY+roiH), (128,0,128), 1)
22             cv2.imshow("YOLOv8 Inference", annotated_frame)
23             cv2.imshow("YOLOv8 Inference M", mainFrame)
24             if cv2.waitKey(1) & 0xFF == ord("q"):
25                 break
26         else:
27             break
28
29 cap.release()
30 cv2.destroyAllWindows()
31

```

```

import cv2
from ultralytics import YOLO

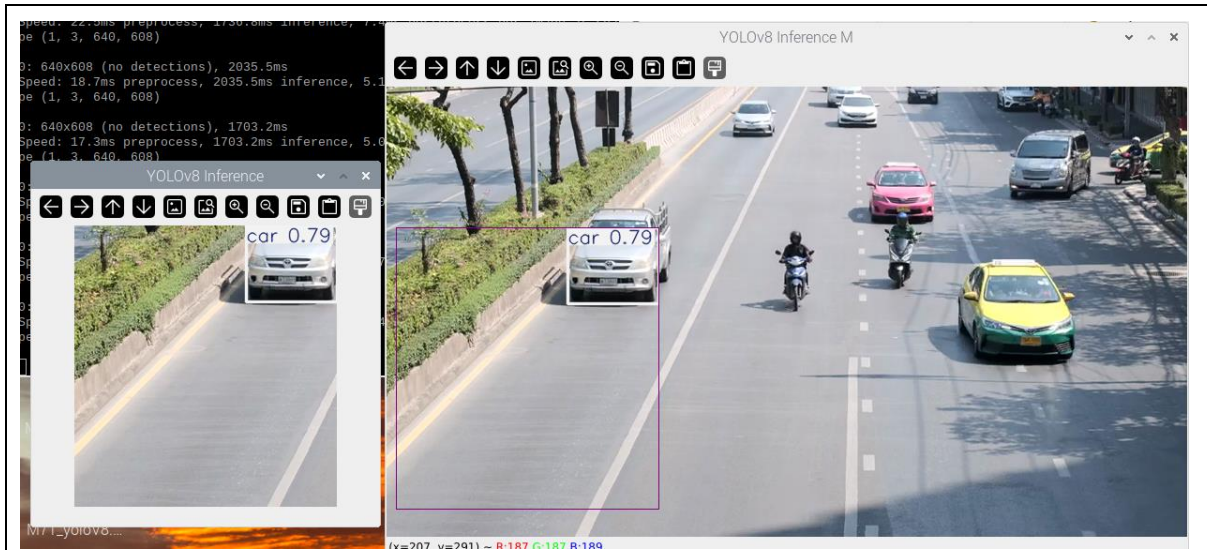
model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt', verbose=False)
video_path = "/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4"

roiX,roiY,roiW,roiH = 10, 150, 280, 300
frame_kStep, frame_count = 8, 0

cap = cv2.VideoCapture(video_path)
while cap.isOpened():
    frame_count += 1
    success, mainFrame = cap.read()
    if success:
        if frame_count % frame_kStep == 0:
            frame = mainFrame[roiY:roiY+roiH, roiX:roiX+roiW]
            results = model(frame, conf=0.55, verbose=True)
            annotated_frame = results[0].plot()
            mainFrame[roiY:roiY+roiH, roiX:roiX+roiW] = annotated_frame
            cv2.rectangle(mainFrame, (roiX,roiY), (roiX+roiW,roiY+roiH), (128,0,128), 1)
            cv2.imshow("YOLOv8 Inference", annotated_frame)
            cv2.imshow("YOLOv8 Inference M", mainFrame)
            if cv2.waitKey(1) & 0xFF == ord("q"):
                break
        else:
            break

cap.release()
cv2.destroyAllWindows()

```

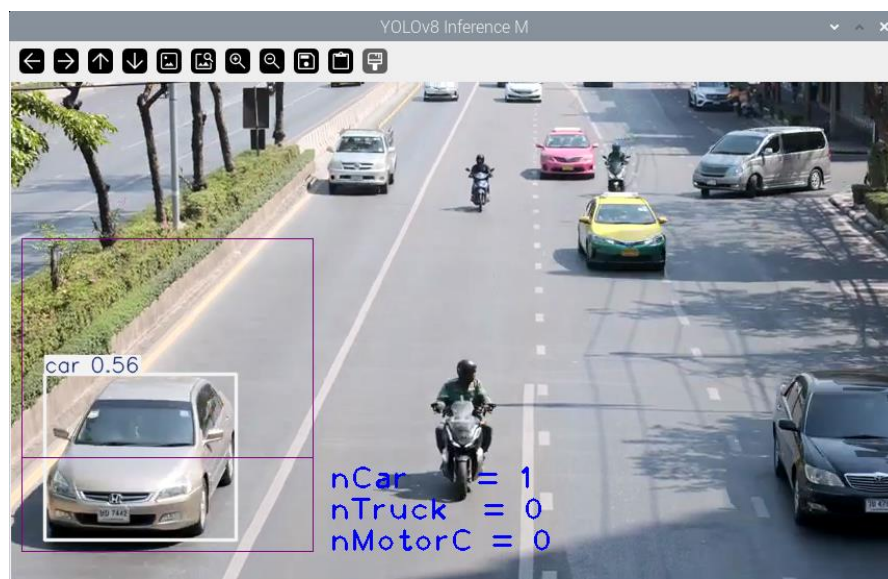


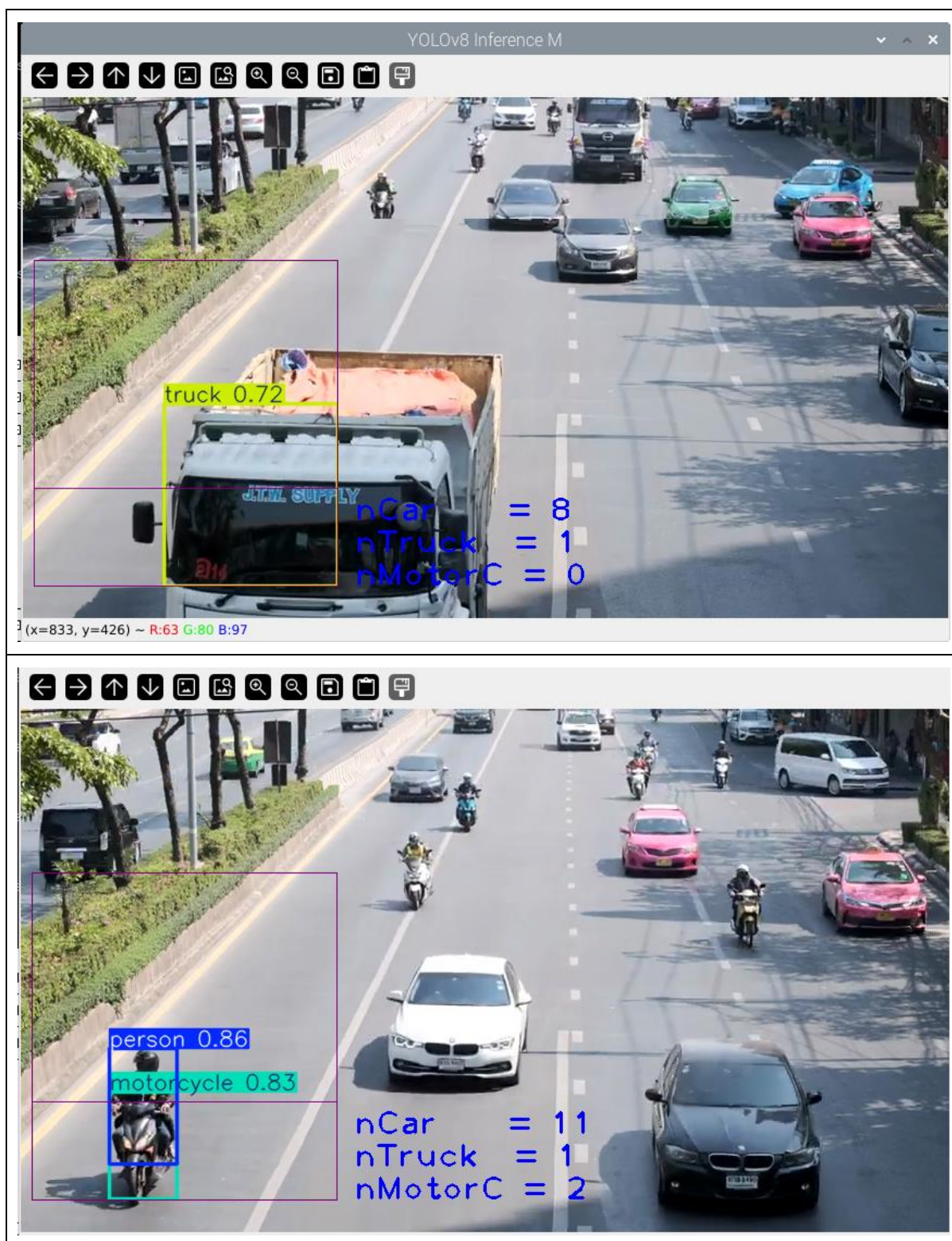
6.3.7.2 -- Count and classify with yoloV8

```

M72_yoloV8_Count.py x
1
2 import cv2
3 from ultralytics import YOLO
4
5 model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt')
6 video_path = "/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4"
7
8 roiX,roiY,roiW,roiH = 10, 150, 280, 300
9 lineY = int(0.8*(roiY+roiH))
10 nCar, nTruck, nMotorC = 0,0,0
11 font = cv2.FONT_HERSHEY_PLAIN
12 frame_kStep, frame_kRepeat, frame_repeat, frame_count = 8, 4, 0, 0
13
14 cap = cv2.VideoCapture(video_path)
15 while cap.isOpened():
16     frame_count += 1
17     success, mainFrame = cap.read()
18     if success:
19         if frame_count % frame_kStep == 0:
20             if frame_repeat > 0:
21                 frame_repeat -= 1
22             frame = mainFrame[roiY:roiY+roiH, roiX:roiX+roiW]
23             results = model(frame, conf=0.55, verbose=True)
24             annotated_frame = results[0].plot()
25
26             for r in results:
27                 boxes = r.boxes
28                 for box in boxes:
29                     cls = int(box.cls[0])
30                     x1, y1, x2, y2 = box.xyxy[0]
31                     x1, y1, x2, y2 = int(x1)+roiX, int(y1)+roiY, int(x2)+roiX, int(y2)+roiY
32                     if (frame_repeat==0) and (y1<lineY) and (y2>lineY):
33                         frame_repeat = frame_kRepeat
34                         if (cls==2): # car id = 2
35                             nCar += 1
36                         if (cls==7): # truck id = 7
37                             nTruck += 1
38                         if (cls==3): # Motorcycle id = 3
39                             nMotorC += 1
40
41             mainFrame[roiY:roiY+roiH, roiX:roiX+roiW] = annotated_frame
42             cv2.rectangle(mainFrame, (roiX,roiY), (roiX+roiW,roiY+roiH), (128,0,128), 1)
43             cv2.line(mainFrame, (roiX,lineY), (roiX+roiW,lineY), (128,0,128), 1)
44             cv2.putText(mainFrame, " nCar = " + str(nCar), (roiX+roiW, roiY+roiH-60), font, 2, (255, 0, 0), 2)
45             cv2.putText(mainFrame, " nTruck = " + str(nTruck), (roiX+roiW, roiY+roiH-30), font, 2, (255, 0, 0), 2)
46             cv2.putText(mainFrame, " nMotorC = " + str(nMotorC), (roiX+roiW, roiY+roiH), font, 2, (255, 0, 0), 2)
47             cv2.imshow("YOLOv8 Inference", annotated_frame)
48             cv2.imshow("YOLOv8 Inference M", mainFrame)
49             if cv2.waitKey(1) & 0xFF == ord("q"):
50                 break
51         else:
52             break
53     cap.release()
54     cv2.destroyAllWindows()
55

```





```

import cv2
from ultralytics import YOLO

model = YOLO('/home/pi/OpenCV_File/data/yolov8n.pt')
video_path = "/home/pi/OpenCV_File/video/ThaiTraffic04_480.mp4"

roiX,roiY,roiW,roiH = 10, 150, 280, 300
lineY = int(0.8*(roiY+roiH))
nCar, nTruck, nMotorC = 0,0,0
font = cv2.FONT_HERSHEY_PLAIN
frame_kStep, frame_kRepeat, frame_repeat, frame_count = 8, 4, 0, 0

cap = cv2.VideoCapture(video_path)
while cap.isOpened():
    frame_count += 1
    success, mainFrame = cap.read()
    if success:
        if frame_count % frame_kStep == 0:
            if frame_repeat > 0:
                frame_repeat -= 1
            frame = mainFrame[roiY:roiY+roiH, roiX:roiX+roiW]
            results = model(frame, conf=0.55, verbose=True)
            annotated_frame = results[0].plot()

            for r in results:
                boxes = r.boxes
                for box in boxes:
                    cls = int(box.cls[0])
                    x1, y1, x2, y2 = box.xyxy[0]
                    x1, y1, x2, y2 = int(x1)+roiX, int(y1)+roiY, int(x2)+roiX, int(y2)+roiY
                    if (frame_repeat==0) and (y1<lineY) and (y2>lineY):
                        frame_repeat = frame_kRepeat
                        if (cls==2): # car id = 2
                            nCar += 1
                        if (cls==7): # truck id = 7
                            nTruck += 1
                        if (cls==3): # Motorcycle id = 3
                            nMotorC += 1

            mainFrame[roiY:roiY+roiH, roiX:roiX+roiW] = annotated_frame
            cv2.rectangle(mainFrame, (roiX,roiY), (roiX+roiW,roiY+roiH), (128,0,128), 1)
            cv2.line(mainFrame, (roiX,lineY), (roiX+roiW,lineY), (128,0,128), 1)
            cv2.putText(mainFrame, " nCar  = " + str(nCar), (roiX+roiW, roiY+roiH-60), font, 2, (255, 0, 0), 2)
            cv2.putText(mainFrame, " nTruck = " + str(nTruck), (roiX+roiW, roiY+roiH-30), font, 2, (255, 0, 0), 2)
            cv2.putText(mainFrame, " nMotorC = " + str(nMotorC), (roiX+roiW, roiY+roiH), font, 2, (255, 0, 0), 2)
            cv2.imshow("YOLOv8 Inference", annotated_frame)
            cv2.imshow("YOLOv8 Inference M", mainFrame)
            if cv2.waitKey(1) & 0xFF == ord("q"):
                break
        else:
            break
cap.release()
cv2.destroyAllWindows()

```